

LESSON PLAN	COURSE: MHF4U
Date: Lesson #8	Unit of Study: Chapter #4 Trigonometry
Materials / Resources: - Advanced Function 12 Study Guide - Chapter #4 & 5 Pre-assessment Quiz - Chapter #4 & 5 Formula, Aid & Homework Sheet(s)	Lesson Focus - Radian Measure - Trigonometric Ratios and Special Angles - Equivalent Trigonometric Expressions
Prior Knowledge: - Basic trigonometry knowledge - Property of a triangle and their characteristics - Trigonometric laws for sine, cosine and tangent	
Overall Expectations: 1. Demonstrate an understanding of the meaning and applications of radian measure 2. Make connections between trigonometric ratios and the graphical and algebraic representations of the corresponding trigonometric functions and between trigonometric functions and their reciprocals, and use these connections to solve problems	
Specific Expectations: 1.1 Recognize the radian as an alternative unit to the degree for angle measurement, define the radian measure of an angle as the length of the arc that subtends this angle at the centre of a unit circle, and develop and apply the relationship between radian and degree measure 1.2 Represent radian measure in terms of π (e.g., $\frac{\pi}{3}$ radians, 2π radians) and as a rational number (e.g., 1.05 radians, 6.28 radians) 1.3 Determine, with technology, the primary trigonometric ratios (i.e., sine, cosine, tangent) and the reciprocal trigonometric ratios (i.e., cosecant, secant, cotangent) of angles expressed in radian measure 1.4 Determine, without technology, the exact values of the primary trigonometric ratios and the reciprocal trigonometric ratios for the special angles $0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$, and their multiples less than or equal to 2π 3.1 Recognize equivalent trigonometric expressions [e.g., by using the angles in a right triangle to recognize that $\sin \sin x$ and $\cos \cos \left(\frac{\pi}{2} - x\right)$ are equivalent; by using transformations to recognize that $\cos \cos \left(x + \frac{\pi}{2}\right)$ and $-\sin \sin x$ are equivalent], and verify equivalence using graphing technology	
Learning Goals: - Students will be able to convert degree measure to radian measure, in exact and approximate formats - Students will be able to convert radian measure to degree measure, in exact and approximate formats	

- Students will know how to determine arc lengths
- Students will be able to determine angular velocity using both degree measure and radian measure
- Students will be able to determine exact values for trigonometric ratios of special angles
- Students will be able to use technology to determine values for trigonometric ratios
- Students will be able to use equivalent trigonometric expressions to simplify calculations

Success Criteria:

1. By the end of this lesson I will be able to convert degree measure to radian measure, in exact and approximate formats
2. By the end of this lesson I will be able to convert radian measure to degree measure, in exact and approximate formats
3. By the end of this lesson I will be able to determine arc lengths
4. By the end of this lesson I will be able to determine angular velocity using both degree measure and radian measure
5. By the end of this lesson I will be able to determine exact values for trigonometric ratios of special angles
6. By the end of this lesson I will be able to use technology to determine values for trigonometric ratios
7. By the end of this lesson I will be able to use equivalent trigonometric expressions to simplify calculations

Assessment Strategies:

Diagnostic: chapter #4 & 5 pre-assessment quiz

Formative: homework for chapters 4.1, 4.2 & 4.3

Summative: quiz #4, test #3

Time:	Introduction:
Part 1	Students will be given a pre-assessment quiz to determine their knowledge standings on trigonometry, and then proceed from their answers.
Time:	Lesson:
Part 2	<u>Chapter # 4.1 Radian Measure</u> Formulate radian measure of an angle and its definition. Give details on the relationships between the radius, arc, and angle. Provide the formulas on converting between degrees measure and radian measure (e.g., from degrees $\rightarrow$ radian, multiply by $\frac{\pi}{180}$ and from radians $\rightarrow$ degrees, multiple by $\left(\frac{180}{\pi}\right)^\circ$). <u>Chapter # 4.2 Trigonometric Ratios and Special Angles</u> Define $\csc x$, $\sec x$, and $\cot x$ and evaluate of these ratios in terms of $\sin x$, $\cos x$, and $\tan x$. Demonstrate the properties of special triangles ($45^\circ - 45^\circ - 90^\circ$ triangle and $30^\circ - 60^\circ - 90^\circ$ triangle) and acknowledge that both are <i>right angle triangles</i>. Demonstrate on a unit circle that $\sin x$ is the value of the

	length that's opposite to the angle θ (hence the y value) and $\cos \cos x$ is the value of the length that's adjacent to the angle θ (hence the x value). Also define the C.A.S.T. rule and what it means. <u>Chapter 4.3 Equivalent Trigonometric Expressions</u> Provide the formulas for co-function identities and use tools such as the unit circle and special angles to prove each statement.
Time: Part 3	Wrap-up: Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions.
Notes, reminders & homework: <u>Reminder</u> - Quiz #4 (to be written before / after lesson #10) <u>Homework:</u> Chapter 4.1: # Chapter 4.2: # Chapter 4.3: #	
Lesson Reflection:	

